Find the first wrong step and repair it

LEARNING TARGET I can use a picture to identify a mistake and explain a correction.

A · KEY VOCABULARY

Quantity	Name what is found: position, speed, etc.
Sign	Direction uses signs; speed is a size.
Force	Sum all forces on the same object.
Check	Use a graph or another equation.

B · KEY EQUATIONS + UNITS

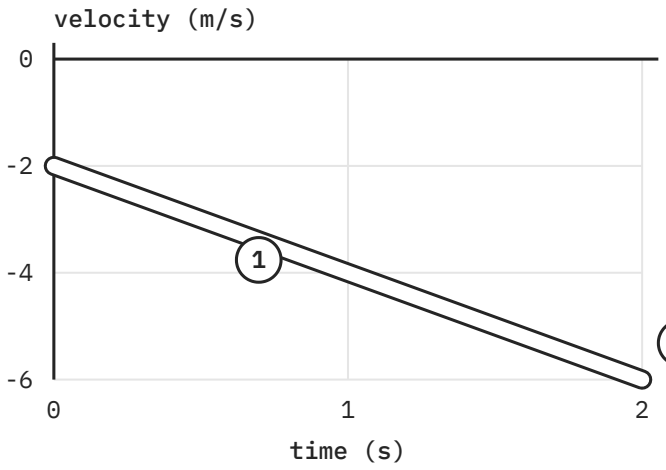
$$\Sigma F = ma$$

$$\Delta v = v_f - v_i$$

units N/kg = m/s²
repair first wrong step

C · COLOR IT · CHECK THE CHANGE IN v

*–2 to –**6 m/s** in **2 s**.*



~~Wrong: $\Delta v = +4$~~ (3)

$\Delta v = -6 - (-2)$

$= -4$ m/s

(2) speed: 2 → 6 m/s

☐ **1 · PURPLE** Velocity line

☐ **2 · GREEN** Increasing speed

☐ **3 · PENCIL** Incorrect Δv

D · WORKED EXAMPLE

A **3 kg** cart has **9 N** right and **3 N** left. Someone writes $a = 9/3 = 3$ **m/s²**. **Explain and repair the first wrong step.**

GIVEN	EQUATION	WORK	ANSWER
m = **3 kg** right is + forces: +**9 N** and –**3 N**	Use net force, not the largest force.	$\Sigma F_x = 9 - 3 = +6$ **N** $a_x = 6/3 = +2$ **m/s²**	+**2 m/s²** Check: ma = **6 N**.

E · THE 2026-XJ THREAD

DAY 19 OF 22 · NEW TOOL: REASONING CHECKS

Before the fictional XJ forecast is shared, check its signs, units, starting coordinates and force sums. Repairing the first wrong step can change every later result.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else